

Kenya Inflation & Commodity Price Analysis

2016 – 2025

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Data Source: Kenya National Bureau of Statistics (KNBS)
Tools: Python (pandas, matplotlib, seaborn, scikit-learn)

1. Overview

This report presents a data-driven analysis of Kenya's Consumer Price Index (CPI) and inflation trends from 2016 to 2025, using official data compiled from Kenya National Bureau of Statistics (KNBS) monthly CPI reports. The analysis examines overall inflation trends, commodity price movements across key sectors including fuel and staple foods, and applies a Linear Regression model to forecast future inflation rates through 2028. The base period for CPI is February 2019 = 100.

2. Dataset Description

The dataset was manually compiled from official KNBS monthly CPI reports covering 10 years (2016–2025). It contains annual average retail prices for selected commodities alongside the overall CPI and inflation rate. The dataset includes 19 variables covering food staples, fuel, electricity, and housing costs. Some commodity columns contain missing values (NaN) for years where data was not available — these were preserved as NaN to maintain data integrity rather than being imputed with estimated figures.

Variable	Description	Unit
Overall Rate of Inflation	Year-on-year inflation rate	%
Overall CPI	Consumer Price Index	Index (Feb 2019=100)
Petrol	Average retail price of petrol	KES per litre
Diesel	Average retail price of diesel	KES per litre
Sugar	Average retail price of sugar	KES per 2 kg
Onions	Average retail price of onions	KES per kg
Tomatoes	Average retail price of tomatoes	KES per kg
Kale (Sukuma Wiki)	Average retail price of kale	KES per kg

Table 1: Key variables in the dataset

3. Key Findings

3.1 Inflation Trends

Kenya's average annual inflation rate between 2016 and 2025 was 6.24%, reflecting a decade of moderate but persistent price pressures. Inflation peaked at 8.3% in 2022 — the highest point in the review period — driven primarily by global supply chain disruptions, the Russia-Ukraine conflict's impact on fuel and food commodity prices, and a weakening Kenya shilling against the US dollar. The Central Bank of Kenya (CBK) responded with successive rate hikes, which contributed to a significant cooling of inflation to its lowest point of 4.3% in 2024, reflecting stabilising fuel prices and tighter monetary policy. The 5% CBK inflation target was breached in most years between 2016 and 2023, underscoring the structural inflationary pressures facing the Kenyan economy.

Metric	Value	Year
Highest Inflation Rate	8.3%	2022
Lowest Inflation Rate	4.3%	2024
Average Inflation Rate	6.24%	2016–2025
Petrol Price Growth	99.7%	2016–2025
Diesel Price Growth	105.2%	2016–2025
Sugar Price Growth	61.8%	2016–2025

Table 2: Summary of key insights

3.2 Fuel Price Trends

Fuel prices showed the most dramatic growth of any category in the dataset. Petrol prices rose by 99.7% between 2016 and 2025 — effectively doubling — while diesel prices grew even faster at 105.2% over the same period. The sharpest increases occurred between 2021 and 2023, coinciding with global oil price surges following the post-COVID demand recovery and geopolitical tensions. Fuel price movements are particularly significant in the Kenyan context because they directly influence transport costs, food distribution expenses, and electricity generation — creating a ripple effect across all other price categories.

3.3 Staple Food Price Trends

Among the staple foods analysed, sugar recorded the highest price growth at 61.8% over the decade, driven partly by global commodity price pressures and domestic supply constraints. Onion and tomato prices showed significant year-to-year variability, reflecting the seasonal nature of horticultural supply in Kenya. Kale (Sukuma Wiki) remained the most consistently affordable staple throughout the period, maintaining its role as a key food security crop for low-income households. The sustained rise in staple food prices has significant implications for household welfare, particularly for women-headed households and low-income urban families who allocate the largest share of income to food expenditure.

4. Inflation Forecast (2026–2028)

A Linear Regression model was applied to forecast Kenya's inflation rate for the next three years, using Year as the independent variable and the Overall Rate of Inflation as the dependent variable.

Year	Predicted Inflation Rate
2026	6.51%

2027	6.55%
2028	6.60%

Table 3: Predicted inflation rates (Linear Regression)

Model Limitation

The model returned an R-squared score of 0.010, indicating a very weak linear relationship between Year and Inflation Rate. This means the model has limited predictive power, and the forecasts should be interpreted with caution. Kenya's inflation is driven by external shocks — including fuel price changes, exchange rate fluctuations, drought, and global commodity trends — that a simple linear model cannot adequately capture. More robust forecasting would require time-series models such as ARIMA or SARIMA that account for seasonality, autocorrelation, and economic volatility. This limitation is acknowledged transparently as part of responsible data practice.

5. Conclusion

This analysis demonstrates that Kenya's inflation over the past decade has been shaped primarily by fuel price volatility and food commodity pressures, with the 2022 peak representing the most severe inflationary episode in the review period. The subsequent decline to 4.3% in 2024 signals a degree of macroeconomic stabilisation, though structural vulnerabilities — including fuel import dependency and climate-sensitive food production — mean that inflation risk remains. The findings have direct relevance to household welfare analysis, particularly for understanding the economic pressures faced by low-income women and families who are most exposed to food and energy price shocks.